

Chapter 47 — Functional Disorders of the Bladder and Bowel in Children

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NORMAL BLADDER AND BOWEL FUNCTION

Lower Urinary Tract Function and Development

- **Mature lower urinary tract (LUT) function is two processes: storage** — filling with normal compliance, no detrusor overactivity, outlet closed at rest; and **voluntary emptying** — coordinated detrusor contraction with lowered resistance of the smooth and striated sphincters.
- Both depend on integration with the **pelvic floor striated muscle, peripheral nervous system and central nervous system (CNS)**; developmental physiology frames childhood **bladder and bowel dysfunction (BBD)**.

Prenatal development

Bladder arises from the urogenital sinus (a cloacal derivative), weeks 4–6 → scattered muscle cells by week 9 → wall thickens through gestation → fetal urine production from the end of the first trimester → **bladder visible on ultrasound at 14 weeks** (storage function has begun).

Animal fetal cystometry

- Sheep: **contractility develops first**; functional innervation continues after muscle development is complete.
- Male fetal pigs: mid-second trimester **constant dribbling** → early third trimester **staccato flow with raised voiding pressure** (detrusor–sphincter discoordination) → detrusor pressure at maximum flow **falls towards term**.

Infancy

- Evidence comes from longitudinal voiding observation and natural-filling cystometry via suprapubic catheters placed at pyeloplasty or nephrectomy. **Interrupted voiding is common**; cystometry shows a rapid rise in detrusor pressure with abrupt cessation of flow.
- **Voiding detrusor pressure frequently exceeds 100 cm H₂O**, more often in boys than girls.
- Several voids within a 10-minute interval; these **frequent voids almost disappear by 2 years** → coordinated voiding is part of normal maturation.
- Capacity variable; postvoid residual (PVR) may be high in an individual void, but **healthy infants almost always empty efficiently**.
- With age frequency falls and capacity rises; the **capacity increase is most marked after 2 years**.

Voiding and sleep

- **Voiding during sleep is uncommon**; with polysomnography no infant voided asleep.
- Sequence: **electroencephalogram (EEG) change** → **infant wakes** → **voids** → **returns to sleep** → higher CNS centres control voiding **before toilet training**.

Concept — *Infant dyscoordination is a transient maturational phase, and cortical arousal gates voiding from birth; dyscoordination that persists beyond toddlerhood is therefore dysfunction, not immaturity.*

Lower Gastrointestinal Function

- **Fetal defecation occurs in healthy pregnancies** (high-resolution ultrasound); it is no longer regarded as a sequela of hypoxia or infection.

Embryology

Primitive gut tube (foregut, midgut, hindgut) forms in **week 4** → blind hindgut ends at the cloacal membrane; distal hindgut expands as the cloaca → **cloaca partitioned into urogenital sinus and anorectal canal, weeks 4–6** → **cloacal membrane ruptures at week 8** → distal third of the anorectal canal develops from the external anal pit.

- **Meconium:** water, intestinal epithelial cells, exfoliated skin cells and hair, gastrointestinal secretions.

Postnatal stool pattern

- Defecation depends on age, diet, intrinsic motility, innervation and comorbidity.
- Avon Longitudinal Study of Parents and Children (ALSPAC): **stool frequency declines through early childhood.**
- **Breastfed infants may pass infrequent but soft stools** — normal.
- Consistency reflects the solid-to-fluid ratio and **correlates with whole-gut transit time**; most newborns pass soft or liquid stool; stools become formed as colonic water conservation matures, and **transit time lengthens from infancy to after puberty.**

Storage and emptying

- Colon propels contents by coordinated smooth-muscle contraction and absorbs water and electrolytes → stool deposited in the rectum.
- **The rectum, like the bladder, is a cloacal derivative and a storage–elimination organ:** highly compliant, smooth muscle quiescent until capacity.

Pre-expulsive phase: rectal distension → **conscious desire via rectal mechanoreceptors and pelvic floor sensory receptors** → **recto-anal inhibitory reflex (RAIR; anal sampling reflex):** internal sphincter relaxes so anal-canal receptors sample contents → **external sphincter contracts simultaneously** to preserve continence.

- **An intact sampling reflex is required for voluntary flatus without faecal incontinence.**

INTERNAL VERSUS EXTERNAL ANAL SPHINCTER

	Internal anal sphincter	External anal sphincter
Muscle	Smooth — thickened extension of colonic smooth muscle	Striated , layered
Control	Autonomic, not voluntary	Spinal and cortical
At expulsion	Relaxes	Relaxes, with pelvic floor

- **Deferral:** external sphincter and pelvic floor contraction; **retrograde motility** holds rectal contents back; rectum stays quiescent until a better opportunity arises or antegrade pressure overcomes it.
- **Expulsive phase:** rectal pressure rises (Valsalva plus propulsive rectal contractions) → coordinated relaxation of internal sphincter, external sphincter and pelvic floor; **squatting or hip flexion straightens the anorectal angle** and eases expulsion.

Concept — *Bladder and rectum are sibling cloacal derivatives running one programme — compliant storage behind a closed outlet, then voluntary emptying through a relaxed outlet — and the functional disorders in this chapter are failures of one of those two phases, usually at the outlet.*

Toilet Training

- Timing is affected by anatomical, physiological and behavioural factors; usual order **bowel control** → **daytime bladder control** → **night-time bladder control.**
- BBD can arise during training; **no consensus on the best method.**

STRUCTURED BEHAVIOURAL VERSUS CHILD-ORIENTED TOILET TRAINING

	Structured behavioural	Child-oriented
Start	Regardless of readiness , from a few weeks of age	On readiness signs
Basis	Caregiver recognises elimination signals — facial change, short cry or grunt; autonomic change (heart rate rises, respiratory rate falls)	Physiological awareness of fullness and ability to delay; cognitive and temperament signs — saying no, broader vocabulary, pride in new skills, putting things where they belong; interest in training, wanting to be clean, indicating wet or dirty pants, pulling clothes up and down
Endorsement	—	American Academy of Pediatrics (AAP)
Outcome	Similar completion; structured approach starts earlier (systematic review)	

Factors affecting age at completion

- **Intrinsic: larger infant bladder capacity persists into childhood and predicts earlier daytime control.**
- **Extrinsic:** mothers working outside the home and ready availability of disposable nappies are suggested causes of later completion; **average age of training has risen worldwide.**
- **Delayed initiation is associated with BBD:** systematic reviews link it to **daytime urinary incontinence and nocturnal enuresis**; Belgian schoolchildren who started later had **more urinary tract infections (UTIs)**.
- **Bowel:** fewer data; later training → more **hiding during defecation**; hiders are at greater risk of **constipation and stool withholding.**

Concept — *Early structured and readiness-based methods reach the same end-point; the harm lies in **late initiation**, which tracks with both urinary and bowel dysfunction.*

Interaction Between the Bladder and Bowel**Physiology**

- Shared embryology, adjacency and parallel storage–elimination function; confinement in the pelvis creates a "**zero-sum**" **volume relationship** in which each organ limits the other's capacity.
- Combined dysfunction in neurological disease (spinal cord injury, spina bifida).
- Healthy adults: **rectal distension alters LUT function and bladder distension alters anorectal function** → translational research.
- **Cross-organ sensitisation:** bladder and bowel afferents run together in the **splanchnic, pelvic and pudendal nerves**; cell bodies in **thoracolumbar and lumbosacral dorsal root ganglia (DRG)** → neural crosstalk lets disease in one organ produce pathology in the other.

Mechanisms in experimental colitis (persist after colitis resolves)

- **Bladder mechanoreceptor hypersensitivity;** altered excitability and channel properties of bladder DRG neurons.
- **A minority of DRG neurons project to both bladder and colon;** separately innervating afferents still **converge on the same spinal cord regions.**
- **Neurogenic inflammation:** efferent neuropeptide release maintains colonic afferent hypersensitivity; neuropeptide expression rises in bladder-innervating DRG and bladder.
- **More activated bladder mast cells and increased urothelial permeability.**

Microbiome

- **Microbiota–gut–brain axis** acts via chemical transmitters, neural pathways and immunity — attractive given the high neurodevelopmental comorbidity of BBD; may also modulate the LUT.

- Urine of healthy individuals contains microbiota; adult urobiome differs in **recurrent UTI, interstitial cystitis/bladder pain syndrome (IC/BPS), urinary incontinence/overactive bladder (UI/OAB) and stones.**
- Children: **pilot case-control study found no urobiome difference in BBD versus controls.**

Concomitant Lower Urinary Tract Dysfunction and Bowel Dysfunction

- **Common and frequently coexistent in healthy children.** Dutch population study (validated questionnaires, toilet-trained children to 7 years): faecal incontinence common and urinary incontinence commoner still, mostly daytime-only; likewise in United States primary care. **Faecal incontinence associated with urinary incontinence in both cohorts.**
- Functional constipation → more **lower urinary tract symptoms (LUTS)**; systematic review: LUTS in a substantial proportion of constipated children.
- **Bowel assessment is indispensable in LUTS:** in tertiary LUTS referrals not on constipation treatment, **the majority met criteria for a functional defecation disorder** — chiefly functional constipation (commonest complaints faecal incontinence and hard or painful stools), plus a further group with **functional non-retentive faecal incontinence (FNRFI).**
- Children with OAB symptoms have more functional constipation and FNRFI.
- Conversely, constipated children **without** LUTS show abnormal pelvic muscle activity during voiding and higher PVR; **rectal distension unpredictably alters bladder capacity, fullness sensation and detrusor overactivity at urodynamics.**

Effects of Treatment: Bowel Treatment on Bladder, Bladder Treatment on Bowel

BIDIRECTIONAL TREATMENT EFFECTS

Direction	Evidence	Verdict
Bowel → bladder	Week-long tailored bowel management programme for severe constipation/faecal incontinence; treating functional defecation disorders	LUTS scores and daytime incontinence improve
Bladder → bowel	European Bladder Dysfunction Study , prospective multicentre	Treating urinary incontinence alone improved functional faecal incontinence
Bladder drug → bowel harm	Anticholinergics, mainstay of OAB treatment	Constipation → return of LUTS after initial improvement if unmanaged
Both	Sacral neuromodulation	Benefits both; literature heterogeneous

KEY POINTS: NORMAL BLADDER AND BOWEL FUNCTION

- Bladder and rectum both store waste and expel it efficiently under voluntary control.
- Physiological and environmental conditions affect toilet-training timing; **delayed training may be associated with BBD.**
- Translational evidence supports a connection between bladder and bowel function.
- LUT and bowel dysfunction **frequently coexist in healthy children.**
- **Untreated bowel dysfunction impairs evaluation and treatment of LUT dysfunction.**
- **Effective treatment of constipation improves LUTS.**

Concept — *Bladder and bowel behave as one functional unit: assess the bowel in every child with LUTS, treat it first, and expect bladder drugs to recreate the bowel problem that drove the symptoms.*

ABNORMAL BLADDER AND BOWEL FUNCTION

- BBD presents with varied LUTS (Table 47.1). **Standardised terminology** is essential for communication and research; the chapter follows the **International Children's Continence Society (ICCS)**.

TABLE 47.1 VARIED LUTS OF BBD

Symptom	ICCS definition
Storage	
Increased frequency	≥8 voids a day
Decreased frequency	≤3 voids a day
Urinary incontinence	Involuntary leakage of urine
Urgency	Immediate need to void
Nocturia	Waking at night to void
Voiding	
Hesitancy	Difficulty initiating the stream when ready to void
Straining	Valsalva needed to initiate and maintain the stream
Intermittency	Non-continuous stream
Urinary retention	Inability to void despite persistent effort, with a distended bladder
Pain	
Dysuria	Burning or discomfort during voiding
Bladder pain	Suprapubic pain or discomfort
Genital pain	Vaginal, penile or scrotal pain
Other	
Holding manoeuvres	Observable strategies to postpone voiding (e.g. crossing legs)
Feeling of incomplete emptying	Bladder does not feel empty after voiding, even if measured PVR is minimal
Recurrent UTIs	Listed as a symptom of BBD

- **Purpose:** BBD is far broader than wetting — storage, voiding, pain and behavioural symptoms all count.
- **Pattern:** frequency is defined by count thresholds either side of normal (≥8 and ≤3 a day).
- **Trap:** sensation of incomplete emptying is a symptom in its own right and does not require a raised PVR; recurrent UTI is itself a presenting symptom of BBD.

Infrequent Urination/Defecation

Urinary Retention

- **Acute urinary retention (AUR)** requires urological consultation and evaluation.
- Consider mechanical obstruction from malignancy and neurological causes, but the **commonest causes are minor genital inflammation or trauma and post-instrumentation**.
- **Almost half occur between the second and fourth year of life;** many non-malignant, non-neurological cases have **acute constipation and faecal impaction**.
- Severe cause not suspected and child not in significant discomfort → **initial approach without catheterisation:** oral analgesia, warm sitz baths and/or enema — less traumatising.
- Hypothesis: a **noxious stimulus such as bowel distension triggers sympathetic dysreflexia** → AUR; urethral catheterisation is a further noxious stimulus.

Constipation

- **Functional constipation:** infrequent, often hard stools; estimated **worldwide prevalence about 14%**; like AUR, **onset peaks at 2–3 years**, often around toilet training.
- Symptoms: infrequent defecation, painful defecation, faecal incontinence, abdominal pain.
- **Organic differential:** congenital anorectal disease; coeliac disease; cystic fibrosis; abnormal abdominal musculature (prune belly, gastroschisis, Down syndrome); neuropathic causes; systemic disease (hypothyroidism, diabetes mellitus); electrolyte disturbance (hypercalcaemia, hypokalaemia); drugs and toxins.
- **Most have no underlying cause → functional (idiopathic) constipation.**

Withholding cycle: painful defecation → voluntary withholding → prolonged retention → more water absorbed → harder stool → more difficult, painful passage → further withholding.

Concept — Retention in a toddler is usually a response to a pelvic noxious stimulus, most often a loaded rectum, so the first manoeuvre is to remove that stimulus rather than add another in the form of a catheter.

Incontinence

Urinary Incontinence

ICCS: DAYTIME URINARY INCONTINENCE VERSUS ENURESIS

	Daytime urinary incontinence (DUI)	Enuresis (bedwetting)
State	Awake — involuntary leakage	Asleep — intermittent leakage
Age	5 years or older	
Note	Treatment varies with aetiology	With or without daytime symptoms

- **Urinary incontinence is associated with constipation and faecal incontinence.**

Faecal Incontinence

RETENTIVE VERSUS NON-RETENTIVE FUNCTIONAL FAECAL INCONTINENCE

	Retentive (majority)	Functional non-retentive (FNRFI)
Mechanism	Overflow: large rectal faecal mass → loss of voluntary control over looser stool accumulating around it	No signs of constipation
Share	Most functional faecal incontinence	About one in five
LUT association	—	DUI and enuresis common, up to half

- **In most children with faecal incontinence no organic cause is found.**

Pain

- Pain is especially distressing for parents; **education and a management plan, once concerning causes are excluded**, allow the family to participate fully.

Dysuria

- Many children referred for "recurrent UTI" have **no positive urine culture**.
- Non-infective sources in BBD: **genital skin irritation from chronic wetting** (topical barrier ointments such as Calmoseptine, nappy ointments); **pain at the end of voiding suggests a bladder source**; bladder irritation.
- Dysuria is non-specific → further evaluation to define the BBD.

- Temporising measures while BBD is treated: **standard urotherapy**, over-the-counter **phenazopyridine**, avoidance of dietary irritants (caffeine, carbonation, red food dyes) — **little paediatric evidence** for dietary restriction.

Lower Abdominal Pain

- Chronic intermittent abdominal pain is common and can accompany BBD; **functional constipation is a common cause of acute and chronic abdominal pain** — the primary cause of about half of acute presentations in one academic primary-care population.

IRRITABLE BOWEL SYNDROME VERSUS FUNCTIONAL DYSPEPSIA

	Irritable bowel syndrome (IBS)	Functional dyspepsia
Features	Abdominal pain with diarrhoea, constipation and/or bloating	Postprandial fullness, nausea, belching
LUT link	Both associated with OAB symptoms, especially nocturia and urgency	

- Question: treat BBD alone, or refer to gastroenterology? Differential divides into organic and functional disorders.
- **Organic causes in prepubertal children:** inflammatory bowel disease, food allergy, coeliac disease, parasitic infection, gastro-oesophageal reflux, acid-peptic disorders.
- **Alarm features** raise the prevalence of organic disease (systematic review by the AAP Subcommittee on Chronic Abdominal Pain and the North American Society for Pediatric Gastroenterology, Hepatology, and Nutrition (NASPGHAN)): **weight loss, gastrointestinal bleeding, persistent fever, chronic severe diarrhoea, significant vomiting** → gastroenterology evaluation.

Genital Pain

- External genital examination usually normal.
- **Referred pain:** in boys, **penile pain is often referred from the bladder** — milder discomfort shows as **grabbing the tip of the penis**; girls report pain vaguely localised to the vaginal area.
- Educating families about referred pain helps, because **treatment is directed away from the site of complaint** — nocturnal perineal pain resolved with relaxation biofeedback for an overactive pelvic floor plus anticholinergics for detrusor overactivity.
- **BBD can cause recurrent epididymitis**; recurrent scrotal pain with bladder outlet dysfunction treated by **pelvic floor therapy** and **onabotulinumtoxinA injection of the external urethral sphincter**.

KEY POINTS: ABNORMAL BLADDER AND BOWEL FUNCTION

- Standardised terminology in BBD is critical for communication and research.
- Common symptoms: **infrequent urination/defecation, incontinence and pain**.
- **Usually no organic cause is identified.**

Concept — Pain in BBD is usually referred or secondary, so once alarm features are excluded, treatment is aimed at the dysfunctional organ rather than the painful site.

ASSOCIATED CONDITIONS

Quality of Life

- Toilet training is developmental and BBD is not life-threatening → **treatment delay is common**; families are told to wait for the child to outgrow it.
- **Consensus across countries:** **nocturnal enuresis, DUI, constipation, faecal incontinence and BBD impair quality of life**; impact greater with **older age, faecal incontinence and combined BBD**.

- Pharmacological treatment → **health-related quality of life improves within 3 months.**
- **Simple non-pharmacological, non-procedural measures also improve quality of life:** randomised controlled trial (RCT) of a group urotherapy session — urinary tract anatomy and physiology, bladder retraining, progressive relaxation, psychosocial aspects — improved quality of life without medication; a 20-page enuresis education booklet with voiding diary improved scores at 1 month versus enuretic controls.

Concept — *Delay is not neutral: quality of life is already impaired, and even education alone improves it, so "wait and outgrow" forfeits an easy gain.*

The Brain and BBD

- **Neurodevelopmental disorders:** early-childhood onset, abnormal brain function, diagnosed through delayed milestones and aberrant behaviour; **exceedingly common** — in a United States cohort of more than 3 million children, commoner among publicly than privately insured children before 8 years.
- **Nocturnal enuresis, DUI, faecal incontinence and constipation are all more prevalent in neurodevelopmental disorders;** three groups carry the highest BBD rates: **attention-deficit/hyperactivity disorder (ADHD), autism spectrum disorder (ASD) and intellectual disability (ID).**
- ICCS consensus non-systematic review on treating incontinence in these disorders.

ADHD, ASD AND ID IN BBD

	ADHD	ASD	ID
Core features	Diagnosable from 3 years by a trained mental health professional; inattentive, hyperactive-impulsive or combined; comorbid oppositional defiant disorder, conduct disorder, depression, anxiety	Deficits in social communication; repetitive patterns of behaviour	Many aetiologies; deficits of varying severity in intellect, attention, communication, memory, motor skills, motivation
BBD link	ALSPAC: ADHD commoner in DUI, monosymptomatic enuresis and faecal incontinence than in continent children; DUI association persists after adjustment for developmental delay and socioeconomic factors	DUI, enuresis, functional constipation and faecal incontinence all commoner	High BBD rates
Drug issues	Stimulants, selective noradrenaline reuptake inhibitors, α 2-adrenergic agonists, antidepressants	No ASD-specific drug; risperidone and other antipsychotics cause urinary and faecal incontinence → time incontinence onset against medication changes	—
Treatment effect	BBD harder to treat, poorer response to behavioural measures, worse outcomes; ADHD treatment alone can improve enuresis and treated ADHD brings continence rates to non-ADHD levels; no effect on functional constipation	Extremely challenging; do not overlook easily reversible causes	Highly individualised
Urotherapy adaptation	Smaller units of information, gradually longer toilet sitting, more teacher/parent supervision, positive feedback	Based on individual behavioural analysis ; liaison with treating psychologist	Multidisciplinary — family, psychologists, teachers, occupational therapy

- **Purpose:** each disorder alters BBD treatment differently — ADHD through attention and drug response, ASD through behaviour and iatrogenic antipsychotic incontinence, ID through global deficits.
- **Trap:** treating ADHD helps the bladder but not the bowel; **bowel management programmes work equally with or without neurodevelopmental disorders** → **ADHD and BBD are assessed and treated in tandem.**

Stressful life events in typically developing children

- Families often ask about psychological causes (death of a pet, moving house, change in family structure, trauma, sexual abuse).
- After sexual abuse or trauma: **education on LUT anatomy and physiology and explicit documentation of a normal examination** reassures caregivers and is often a prerequisite for engaging in urotherapy.
- United Kingdom community cohort: **girls** reporting stressful life events at 7–8 years had more new-onset urinary incontinence at 9 years; **not seen in boys**; causation undetermined.
- Sri Lanka: constipation commoner with stressors such as an alcoholic father or parental unemployment; greater BBD severity at presentation associated with **unplanned pregnancy, daycare attendance and school problems**.

Anxiety and depression

- **Common in BBD; direction of causality uncertain.**
- ALSPAC adolescents: **depressive symptoms** ↔ DUI, faecal incontinence, voiding postponement; **generalised anxiety disorder** ↔ DUI, voiding postponement, enuresis, frequent urination; similar associations in North America, South America and Asia.
- **Urologist's role:** ask about diagnoses and review medications that imply them; if clinic observation suggests an undiagnosed disorder, **liaise with primary care**; treating the comorbidity helps both BBD and overall health.

Concept — Neurodevelopmental and psychological comorbidity is the rule rather than the exception in BBD; it lowers the ceiling on behavioural therapy, so it is identified and treated alongside the bladder and bowel rather than after them.

Associated Conditions in Adulthood

- Mothers (usually the accompanying parent) with postpartum incontinence, IC/BPS or OAB ask whether the child's symptoms predict adult disease.
- Two study designs: **children with BBD followed into adulthood**, or **adults with LUTS recalling childhood symptoms**.
- **Childhood LUTS predict adolescent and adult LUTS** — longitudinal cohort, case-control, cross-sectional follow-up; **more than a quarter of constipated young adults have been symptomatic since childhood**.

Risk factors for persistence or recurrence

- Women with **urge and mixed urinary incontinence** more often report childhood BBD.
- **Higher childhood BBD symptom score** → adult female urinary incontinence and adult OAB, with greater OAB severity and worse quality of life.
- **Prolonged treatment and later symptom recovery:** DUI plus enuresis still present at **9 years** persists into late adolescence; OAB recovery after **12 years** predicts adult OAB.
- **A favourable childhood treatment outcome does not lower adult LUTS risk:** complete, partial and non-responders had equal adult LUTS and incontinence.
- Dysfunctional voiding in girls (RCT 2000–2003 of outpatient behavioural therapy alone, with home video instruction, or with home uroflow biofeedback) → **20 years later uroflow and emptying had improved further, but the proportion with DUI was essentially unchanged**.

Interstitial cystitis/bladder pain syndrome

- Women with IC/BPS recall **more childhood UTIs and urgency** than controls.

- IC/BPS with IBS-like symptoms → higher pelvic pain and urgency/frequency scores and more recalled childhood BBD: no daily bowel movement, pushing to defecate, voiding only once or twice daily, painful urination.
- Implication: BBD has a **longitudinal component warranting transitional care**, as for structural and neurological paediatric urology.

KEY POINTS: ASSOCIATED CONDITIONS

- Conservative non-pharmacological management of BBD **improves quality of life**.
- BBD is **more prevalent in neurodevelopmental disorders**, which may alter the treatment plan.
- Childhood BBD is **associated with persistence of LUTS into adulthood**.

Concept — BBD behaves as a lifelong phenotype rather than a phase: it tracks into adult OAB, incontinence and IC/BPS regardless of how well childhood treatment worked, which argues against delay and for planned transition.

EVALUATION

- Several societies address BBD evaluation and treatment (Table 47.2); the approach below follows them.
- **Diagnosis and planning rest largely on history and examination**; simple tests help, but **children with BBD are often over-investigated**.
- **Most LUTS and bowel dysfunction are functional**; history and examination exist chiefly to **exclude anatomical or organic disease**.
- Warning signs (Table 47.3) prompt evaluation for **chronic kidney disease (CKD)**, **neurogenic causes**, **endocrine causes (diabetes mellitus, diabetes insipidus)** and **non-functional gastrointestinal disorders**.

TABLE 47.2 SOCIETY GUIDELINES THAT ADDRESS BBD

Topic	Society (reference, year)
Terminology and assessment	
Standardisation of terminology	ICCS (Austin et al., 2016)
Diagnostic scores, questionnaires, quality of life	ICCS (Chase et al., 2018)
Urodynamic studies	ICCS (Bauer et al., 2015)
Daytime LUT conditions	
Daytime LUT conditions / DUI	EAU/ESPU (Tekgul et al., 2020); ICCS (Chang et al., 2017)
Urotherapy	ICCS (Nieuwhof-Leppink et al., 2021)
Dysfunctional voiding	ICCS (Chase et al., 2010)
Enuresis	
Enuresis / nocturnal enuresis	EAU/ESPU (Bogaert et al., 2020); ICCS (Nevéus et al., 2020)
Non-monosymptomatic enuresis	ICCS (Franco et al., 2013)
Infection and reflux	
UTI	EAU/ESPU ('t Hoen et al., 2021)
BBD and UTI	ICCS (Yang et al., 2018)
Vesicoureteral reflux (VUR)	AUA (Peters et al., 2017)
Bowel	
Functional constipation	

Topic	Society (reference, year)
	ESPGHAN/NASPGHAN (Tabbers et al., 2014); ICCS (Burgers et al., 2013)
Functional non-retentive faecal incontinence	ICCS (Koppen et al., 2016)
Colonic function investigations	ESPGHAN (Rybak et al., 2022)
Probiotics; synbiotics	ESPGHAN (Szajewska et al., 2023; Hojsak et al., 2023)
Neurodevelopment	
Neurodevelopmental disorders and incontinence	ICCS (von Gontard et al., 2022)

AUA, American Urological Association; EAU/ESPU, European Association of Urology/European Society for Paediatric Urology; ESPGHAN, European Society for Paediatric Gastroenterology, Hepatology and Nutrition.

- **Pattern:** the ICCS covers almost every topic; DUI, enuresis and functional constipation each have **two bodies** — a urological or gastroenterological society plus the ICCS.
- **Trap:** the key bowel guideline for urologists is gastroenterological (ESPGHAN/NASPGHAN 2014), while the "treat the bowel first" rule comes from EAU/ESPU.

TABLE 47.3 BBD WARNING SIGNS IN THE CHILD/ADOLESCENT

Source	Warning sign	Directs evaluation towards
History	Blood in stool without anal fissure; fever, vomiting, weight loss	Organic gastrointestinal or systemic disease
Growth, vital signs	Failure to thrive; raised blood pressure	Organic disease; renal involvement/CKD
Abdomen	Severe abdominal distension	Non-functional bowel disease
Spine	Hair tuft, sacral dimple, gluteal cleft deviation	Occult spinal dysraphism (neurogenic)
Lower limbs	Reduced strength or sensation; gait abnormality	Neurogenic cause
Anus	Abnormal position; absent anal or cremasteric reflex	Anorectal malformation; neurogenic cause
Urinalysis (UA)	Low specific gravity, glucosuria, proteinuria	Diabetes insipidus/CKD; diabetes mellitus; CKD

- **Purpose:** a checklist that licenses investigation — without these signs, BBD is treated as functional.
- **Trap:** the spine, lower limbs and anal and cremasteric reflexes are part of every BBD examination, not only of suspected neurogenic cases.

History

- **Barriers:** school-age children toilet independently and spend much of the day away from home → parents lack detail; children misunderstand abstract questions.
- **Ask concretely:** children cannot say whether they are constipated but can report bowel movements per day or week, painful defecation and **stool form from Bristol pictures** (Fig. 47.1); ask how many times something happened in a set period rather than how often.
- Child and parent provide complementary information; the first history sets the **team approach** — **child, parent and clinician**; families may already be frustrated by earlier neglect.
- Families volunteer only the most bothersome symptom (nightly bedwetting eclipses intermittent DUI at play; urology clinics hear about LUTS over bowel) → **assess bowel function in every child with LUTS**, even unprompted.
- **Assess bother to the child** — an unbothered child is hard to engage in behavioural therapy.

BLADDER VERSUS BOWEL DIARY

	3-day bladder diary	3-day bowel diary
Records	LUTS, frequency and volume of voids , incontinent episodes, fluid intake	Defecation frequency, stool form on Bristol chart , incontinent episodes, diet, fluid intake

- **Prior treatment in detail:** often given without proper instruction — **maintenance laxatives without prior clean-out; enuresis alarm expected to work immediately** → re-education on why results will differ before retrying.
- **Past medical history:** neurodevelopmental diagnosis stable or still being adjusted; input from the family's mental health professional; systemic disease to optimise before anything beyond standard urotherapy (e.g. poorly controlled diabetes).
- **History that may affect LUT innervation raises suspicion of neurogenic bladder/bowel even without visible deficit:** tethered cord, low-lying conus medullaris/fatty filum terminale, transverse myelitis, pelvic surgery.
- **Family history:** little evidence that BBD is genetic, but **mothers with BBD are more likely to have a child with BBD**; relatives' inaccurate expectations from their own treatment limit their role as caregivers → education.
- **Social and developmental history:** disruption at school and home; brief functional screen is appropriate, formal diagnosis is not the urologist's. **Strengths and Difficulties Questionnaire (SDQ):** beyond those with known diagnoses, a further subset without prior diagnosis scored in the clinical range → **usable as a behavioural screen in BBD**.
- Behavioural interventions need **consistency** — identify barriers where the child is at school most of the day or lives in more than one household.

FIG. 47.1 BRISTOL STOOL CHART (legend)

Type	Description
1	Separate hard lumps, like nuts (hard to pass)
2	Sausage-shaped but lumpy
3	Like a sausage but with cracks on the surface
4	Like a sausage or snake, smooth and soft
5	Soft blobs with clear-cut edges
6	Fluffy pieces with ragged edges, a mushy stool
7	Watery, no solid pieces; entirely liquid

Physical Examination

- Explain purpose and scope to parents and, if old enough, the child.
- **AAP chaperone policy:** infant or child — **always with parent or guardian present**; chaperone if unavailable or if their presence interferes; adolescent — **chaperone recommended for genital examination**.
- **General:** abnormal growth parameters, particularly with bowel dysfunction → organic or anatomical cause; behaviour during history → need for neurodevelopmental screening, or infeasibility of attention-dependent therapy such as pelvic floor biofeedback; **watch the child move and climb onto the couch** for lower-limb motor deficit or gait disturbance.
- **Spina bifida is the commonest cause of neurogenic bladder in children.** Markers of occult/closed spina bifida: **midline lumbosacral cutaneous lesion** (dimple, hair tuft, dermal vascular malformation) or **asymmetric gluteal cleft** → **low threshold for spinal imaging**, also in intractable BBD.

- **Abdomen:** core muscle tone, palpable bladder, faecal mass, tenderness.

Anogenital examination

- Urine or stool on underwear; skin breakdown from urine or stool exposure.

WETTING SIGNS AT EXAMINATION IN GIRLS

Finding	Suggests
Gush of urine when the labia majora are spread	Vaginal trapping (urethrovaginal reflux)
Constant leakage during examination	Ectopic ureter

- Anus: position; fissure; **skin tags — hallmark of a chronic healing fissure**; fissures are exquisitely painful and may limit examination; **anterior displacement or small calibre → possible anorectal malformation**.
- **Digital rectal examination (DRE)** can assess anal stenosis or sphincter tone, but **evidence does not support routine DRE to diagnose functional constipation**.

Evaluation of Lower Urinary Tract Function

Validated Questionnaires and Semiquantitative Assessment

- LUTS are largely subjective → questionnaires quantify severity and monitor response.

VALIDATED PAEDIATRIC SYMPTOM SCORES

Instrument	Items	Scope	Note
Pediatric Urinary Incontinence Quality of Life score (PIN-Q)	28	LUTS only (quality of life)	Cross-culturally reliable
Dysfunctional Voiding Symptom Score (DVSS)	10	LUTS + one bowel item ("I miss having a bowel movement every day")	Cross-culturally reliable; bowel item flags healthy children without daily stools
Vancouver Symptom Score	14	Validated for BBD; three bowel items — frequency, consistency, faecal incontinence	One instrument covers both systems in a busy clinic

ICCS TREATMENT RESPONSE DEFINITIONS

Response	Reduction in symptoms
Complete	100%
Partial	50–99%
None	<50%

Record the initial response and the response at a fixed time after treatment stops — critical for research.

Urinalysis

- **Critical component, but false positives are common**; costly and invasive work-ups from false positives led the **AAP to remove routine screening urinalysis in asymptomatic children in 2007**.

URINALYSIS FINDINGS AND ACTION IN BBD

Finding	Concern	Action
Low specific gravity (with thirst, poor growth, dehydration)	Diabetes insipidus (rare cause of frequency/incontinence); specific gravity >1.005 makes it unlikely	Serum electrolytes and glucose, plasma osmolality, urine osmolality, plasma antidiuretic hormone (ADH)
Glucosuria	Diabetes mellitus	Blood glucose
Persistent proteinuria or suggestive history	CKD (impaired concentrating ability)	Renal ultrasound, serum electrolytes
	Usually benign	Observe; repeat urinalysis

Finding	Concern	Action
Isolated microscopic haematuria, no suggestive history (UTI, stones, trauma)		
Isolated dipstick proteinuria	Rarely significant	If $\geq 1+$ persists after standard urotherapy → first-morning urine protein:creatinine ratio; >0.2 mg/mg → further evaluation
Pyuria, positive culture	Contamination common	Treat if LUTS recent-onset, acutely worse or febrile; otherwise decide on symptoms

- **Contamination:** clean-catch specimens are frequently contaminated in children older than 3 years; **vulvar irritation from wet underwear causes false-positive pyuria and haematuria** in girls.
- **Trap:** positive dipstick blood or protein is frequent in BBD and does not by itself justify investigation.

Postvoid Residual/Ultrasound

- **Pelvic ultrasound is a key non-invasive tool;** bladder scanners assess completeness of emptying, but **PVR in healthy children is variable → standardise conditions:**
 - bladder **not under-distended** ($<50\%$ expected bladder capacity, EBC) or **over-distended** ($>115\%$ EBC);
 - measure **within 5 minutes of voiding;**
 - ICCS abnormal values depend on **single versus repeated measurement and on age.**
- **B-mode ultrasound** also measures PVR and shows the bladder and surroundings, but the extra information **largely does not change management;** thickened bladder wall is frequent and is associated with hypocontractile bladder function.
- **Renal ultrasound is generally not indicated in BBD;** however, **DVSS ≥ 12 plus a history of UTI** predicts abnormal renal-bladder ultrasound.

Uroflowmetry

- **First-line testing for suspected LUT dysfunction: PVR and uroflowmetry.**
- Normal voiding: outlet relaxes, detrusor contracts against minimal urethral resistance → **bell-shaped curve.**
- **Uroflowmetry alone cannot rule dysfunction in or out** — abnormal curves occur in healthy children and vary between voids → **ideally three flows.**
- Conditions: void as normally as possible; boys aim at a fixed point in the funnel; **voided volume $>50\%$ EBC for age or greater than the maximum voided volume on the diary.**
- **ICCS interpretation:** curve shape (**bell, interrupted, plateau, staccato, tower**), maximum flow (Q_{max}), average flow (Q_{ave}).
- Refinements for inter-rater reliability: **flow index** = actual flow rate / flow rate estimated from a bell curve, from equations derived in normally voiding children; **Societies for Pediatric Urology (SPU) Voiding Dysfunction task force:** shape **smooth or fractionated** — fractionated = **pause ≥ 1 second** and/or **drop in velocity exceeding the square root of the preceding Q_{max} ;** clinical correlations need study.

Electromyography

- **Electromyography (EMG)** complements curve shape and flow rate; in fractionated curves, pelvic floor and abdominal EMG distinguish an **intermittently relaxing sphincter from abdominal straining.**
- **Active pelvic floor EMG during voiding → dysfunctional voiding.**
- **EMG lag time** (pelvic floor quieting → onset of flow) **<2 seconds** → detrusor already contracting as the sphincter releases → **detrusor overactivity.**

Urodynamics

- **Invasive urodynamics are not routine in neurologically intact children.**
- **Failed treatment → re-evaluation, which may include invasive urodynamics; add fluoroscopy to assess outflow obstruction.**

Concept — Lower urinary tract evaluation is deliberately non-invasive and stepwise — diary, urinalysis, PVR, flow with EMG — and invasive urodynamics are earned by treatment failure, not by the presenting symptom.

Evaluation of Bowel Function

- **Functional constipation is diagnosed on history and examination;** further investigation adds nothing routinely but may help when constipation is **refractory**.

Diagnostic Criteria

- **Rome IV criteria** are the most widely accepted definition; criteria depend on **age and toilet-training status**.

TABLE 47.4 ROME IV CRITERIA FOR FUNCTIONAL CONSTIPATION IN A CHILD/ADOLESCENT

At least two of the following, at least once a week for at least 1 month, and insufficient criteria for IBS:

- **Two or fewer defecations in the toilet per week** (developmental age at least 4 years).
- **At least one episode of faecal incontinence per week.**
- Retentive posturing or excessive volitional stool retention.
- Painful or hard bowel movements.
- Large faecal mass in the rectum.
- Large-diameter stools that can obstruct the toilet.
- **Trap: reduced frequency alone is insufficient; abdominal pain is not a criterion;** functional constipation cannot be diagnosed if IBS criteria are met.

FUNCTIONAL CONSTIPATION VERSUS IBS

	Functional constipation	IBS
Abdominal pain	Frequent but not a criterion	Pain associated with defecation
Stool	Rome IV features above	Change in frequency and/or appearance
When constipation resolves	Pain settles	Pain persists

- **Modified Bristol stool form scale** (Fig. 47.1): commonest paediatric consistency tool; **reliable and valid from 6 years when descriptors are read aloud;** caregivers often do not see every stool of a toilet-trained child.
- **ESPGHAN/NASPGHAN:** diagnosis primarily on history and examination; **no routine laboratory testing** without suggestive findings; **hypothyroidism, coeliac disease, hypercalcaemia and cow's milk allergy are not routinely screened.**

Radiologic Studies

- **Plain abdominal radiograph:** often ordered for stool burden, but **neither sensitive nor specific**, with poor inter- and intra-observer reliability and no correlation with symptoms (systematic review) → **guideline recommends against it for diagnosis;** permitted if impaction is suspected and examination is impossible or unreliable; useful for stool burden in **refractory** cases.

RECTAL ULTRASOUND FOR CONSTIPATION: THE TWO GUIDELINES

EAU-ESPU	ESPGHAN/NASPGHAN
Rectal diameter ≥ 30 mm on transabdominal sonography may indicate constipation	No routine ultrasound to diagnose functional constipation

Constipated children are more likely to have a wide rectum, but any cut-off misclassifies some children; ultrasound's value for bowel assessment is less clear than for the bladder.

Additional Studies

- Not generally indicated without signs or symptoms; the urologist treating constipation needs basic familiarity with them.

ADDITIONAL BOWEL INVESTIGATIONS

Test	What it shows	Role
Water-soluble contrast enema	Colonic position, redundancy, areas of chronic distension, stool burden	Helps interpret plain-film stool appearance and so drive therapy
Colonic transit study	Colonic transit time (prolonged in constipation); capsule of radio-opaque markers + serial radiographs; scintigraphy an alternative	When management proves challenging; capsule method cheap and office-based
Anorectal manometry	Sphincter function, anorectal sensation, rectoanal reflexes, simulated defecation; 2D pressures or 3D topography	Main indication: test the RAIR in older children with constipation suggestive of Hirschsprung disease (e.g. early onset) → absent RAIR → rectal biopsy, the criterion standard
Colonic manometry	Intraluminal pressures and coordination — normal physiology versus neuromuscular disease; high-amplitude propagating contractions , responses to a meal and stimulants (bisacodyl, glycerin)	3-day admission ; bowel cleansing, probe placed under general anaesthesia at colonoscopy; limited to large neurogastroenterology/motility centres; option before surgery for intractable constipation

KEY POINTS: EVALUATION

- Warning signs prompt evaluation for **non-functional diagnoses**.
- Focused questions overcome barriers to a complete BBD history.
- Examination focuses on **development and the back, abdomen and genitals**.
- LUTS evaluation: **urinalysis, PVR and uroflowmetry** — not usually urodynamics.
- **History and examination alone usually diagnose functional constipation.**

Concept — Bowel dysfunction is diagnosed clinically by Rome IV; imaging and physiology are held back for refractory disease, where their job is to find Hirschsprung disease and to map colonic motility before any operation.

URINARY INCONTINENCE: DAYTIME URINARY INCONTINENCE

- Urinary incontinence divides into leakage **while awake (DUI)** and **while asleep (enuresis)**; Fig. 47.2 gives the approach for children older than 5 years.
- **History, examination, urinalysis and PVR** suffice to exclude **ectopic ureter, neurogenic bladder, diabetes mellitus, diabetes insipidus** and other non-functional causes.
- **Screen for constipation; EAU-ESPU: treat bowel function before specific treatment of LUT dysfunction.**

- **Recognise and optimise neurodevelopmental and psychiatric disorders**, which modify the treatment approach.

FIG. 47.2 ALGORITHM FOR TREATMENT OF URINARY INCONTINENCE (legend)

Assess: history → examination → urinalysis, PVR, uroflow.

Exclusion cascade (each "yes" leaves the functional pathway):

- Suspected neurogenic bladder → **urodynamics and magnetic resonance imaging (MRI) of the spine**.
- Suspected diabetes insipidus or mellitus → evaluation and appropriate referral.
- Suspected CKD → **renal ultrasound; rule out obstruction**.
- Suspected anatomical abnormality → evaluation and appropriate treatment.

Functional pathway: screen for neurodevelopmental disorder → optimise its treatment → **constipation?** → **treat it** → **night symptoms only?** → **treat monosymptomatic enuresis** → otherwise **standard urotherapy** → day symptoms resolved? (yes → treat any residual enuresis as monosymptomatic) → no → **specific urotherapy** → resolved? (yes → as before) → no → **consider third-line treatment**.

Standard Urotherapy

- **Non-pharmacological, non-surgical behavioural treatment**, largely built on cognitive behavioural therapy techniques; **first-line treatment for functional LUT dysfunction**.
- Goal: teach the child **how to void, when to void and how often to void**; several European countries require **advanced postgraduate training**.
- Just as "eat less and move more" rarely produces weight loss, **urotherapy is not simply telling children to void more often and avoid constipation**.

Elements (ICCS)

- **Information:** normal LUT function and how the child's differs.
- **Instruction:** how to resolve the dysfunction.
- **Lifestyle advice:** fluid intake, diet, regular bladder and bowel emptying.
- **Registration:** bladder diaries or mobile apps for self-awareness; the diary becomes feedback as treatment progresses.
- **Support and encouragement:** regular follow-up with the caregiver.

Specific instructions

- **Toileting posture:** **hips and knees at about 90°**, with foot stool and/or seat insert as needed → pelvic floor relaxation.
- **Deep diaphragmatic inhalation and slow exhalation** relax the pelvic floor.
- **Regular voiding at least six to seven times a day**; for voiding postponement, anchor voids to events (before meals, start of school recess, before reading).
- School toilet environment is associated with LUTS → agree an achievable school voiding plan.
- **Child owns adherence** (to developmental level), family supports; **operant conditioning rewards collaboration, not dryness** — the child controls adherence, not incontinent episodes; the child fills in the diaries; a **timer watch** prompts toilet visits rather than repeated reminders from adults.

Duration and outcome

- **ICCS: standard urotherapy for 4 weeks**.
- Meta-analysis: **more than half of children with DUI resolve with standard urotherapy, far above the spontaneous cure rate**.

- No improvement → **re-evaluate**; findings after standard urotherapy direct **specific urotherapy**.

Concept — *Standard urotherapy is a structured behavioural treatment with defined elements, a defined duration and rewards tied to adherence rather than outcome; "void more often" is not urotherapy.*

Dysfunctional Voiding

- **Definition:** the **external urethral sphincter–pelvic floor complex fails to relax during voiding in a neurologically intact child**; ICCS — the child **habitually contracts the urethral sphincter or pelvic floor during voiding** and shows a **staccato flow pattern**.

TERMS THAT ARE NOT INTERCHANGEABLE

Dysfunctional voiding	Detrusor sphincter dyssynergia (DSD)
Sphincter–pelvic floor discoordination in a neurologically intact child	The same discoordination in the presence of a neurological condition
Dysfunctional voiding	Voiding dysfunction
A specific voiding-phase diagnosis	Historical umbrella term ; current usage is LUT dysfunction, which covers storage and voiding disorders

- **Aetiology — learned, acquired behaviour:** normal sequence is sphincter–pelvic floor relaxation then detrusor contraction; voluntary pelvic contraction at the desire to void develops to **prevent incontinence, avoid a public toilet, or avoid pain from vulvitis or balanitis**.

Consequence cascade: functional outflow obstruction → difficulty initiating or maintaining the stream, abnormal stream → **proximal urethral dilatation** → bladder: compensatory hypertrophy, detrusor overactivity and/or reduced compliance → detrusor decompensation → emptying needs **clean intermittent catheterisation (CIC)** or surgery → most severe: **secondary VUR, renal deterioration, hypertension**, as described by **Hinman**.

- **Boy with hypertension and urinary incontinence** → **renal ultrasound**; upper-tract dilatation → differential of **anatomical obstruction, dysfunctional voiding, neurogenic LUT dysfunction**.
- **Suspect** with obstructive symptoms — hesitancy, abnormal stream, incomplete emptying; bladder ultrasound: thick wall, diverticula, high PVR.
- **Diagnosis: obstructed voiding pattern on uroflowmetry with active pelvic floor EMG**.
- Further as appropriate: renal ultrasound (upper tract), **spinal MRI** (spinal abnormality), **cystoscopy** (anatomical obstruction).

DYSFUNCTIONAL VOIDING: MANAGEMENT BY RENAL RISK

Significant risk of kidney damage	Low renal risk
Invasive video urodynamics; control bladder storage pressure; emptying that bypasses the sphincter (CIC)	Primary aim: retrain the outlet to work with the bladder — continue standard urotherapy with emphasis on posture and breathing; double voiding if PVR large ; treat constipation and behavioural/psychological issues; short-course antibiotic prophylaxis if recurrent UTI ; a minority normalise on this alone

- **Biofeedback is first-line specific urotherapy for dysfunctional voiding:** meta-analysis — versus standard urotherapy, **fewer UTIs, better flow, fewer abnormal voiding EMGs**. Visual or auditory signals of flow and pelvic floor relaxation; the child watches their own flow curve, and **perineal EMG biofeedback** isolates the pelvic floor.
- **α-Adrenergic antagonists** (lower evidence): **doxazosin** in LUTS with raised PVR — populations not defined by EMG, so not purely dysfunctional voiding; **added after biofeedback failure it reduced PVR**.

- **Intrasphincteric onabotulinumtoxinA** for therapy-resistant cases: reduced PVR, incontinence and UTI; **doses 50–300 units** (systematic search and narrative review).

Concept — *Dysfunctional voiding is a learned outlet behaviour that acts like a mechanical obstruction, so the upper-tract risk decides management: bypass the outlet when the kidneys are threatened, retrain it when they are not.*

Overactive Bladder

- **ICCS definition:** urinary urgency, usually with frequency and nocturia, with or without incontinence, in the absence of UTI or other obvious pathology.
- OAB incontinence is usually urgency incontinence, but **urgency incontinence does not by itself establish OAB**: children with **voiding postponement** also leak with a sudden urge; **treating them as OAB helps at first and later worsens the LUT dysfunction**.

Oral Medications

- Considered **only after standard urotherapy and constipation management**, according to individual goals.
- Adults: **antimuscarinics** and oral **β3-adrenoceptor agonists** are initial pharmacotherapy; **extended-release (ER) preferred over immediate-release (IR)** for fewer side effects.
- Children differ: in the United States **oxybutynin is the only drug approved for paediatric non-neurogenic OAB**; **ER tablets are usually unusable if the child cannot swallow a pill**, because the coating provides the extended release.

TABLE 47.5 MEDICATIONS FOR PEDIATRIC OAB (modified from Ramsay et al., 2022)

Agent	Formulation	Recommended dose
Antimuscarinics		
Oxybutynin	IR suspension 1 mg/mL; IR tablet 5 mg	0.3–0.6 mg/kg/day in 2–3 divided doses (maximum 15 mg/day)
	ER tablet 5, 10, 15 mg	5–20 mg once daily
	Transdermal patch 3.9 mg/day/patch	1–2 patches every 3–4 days
Propiverine	IR tablet 5 mg	<35 kg: 0.8–1.2 mg/kg/day in 2 divided doses (maximum 30 mg/day)
	ER tablet 30, 45 mg	>35 kg: 30–45 mg once daily
Tolterodine	IR tablet 1, 2 mg	1–2 mg twice daily
	ER tablet 2, 4 mg	2–4 mg once daily
Fesoterodine	ER tablet 4, 8 mg	4–8 mg once daily
Trospium	IR tablet 20 mg	20 mg twice daily
	ER tablet 60 mg	60 mg once daily
Solifenacin	ER tablet 5, 10 mg	2.5–10 mg once daily
β3-Agonists		
Mirabegron	ER suspension 8 mg/mL	11–22 kg: 3–6 mL; 22–35 kg: 4–8 mL; >35 kg: 6–10 mL — once daily
	ER tablet 25, 50 mg	>35 kg: 25–50 mg once daily
Vibegron	ER tablet 75 mg	No paediatric dose given

- **Purpose:** weight-banded paediatric dosing; **35 kg is the recurring switch** from weight-based or liquid dosing to adult-style tablets (propiverine, mirabegron).
- **Pattern:** only oxybutynin (suspension) and mirabegron (ER suspension) come as liquids.

- **Trap:** the oxybutynin maximum is 15 mg/day IR, but the ER range extends to 20 mg; vibegron has no paediatric dose.

PAEDIATRIC OAB AGENTS COMPARED

Agent	Receptor profile	Structure / CNS	Paediatric status and key evidence	Signature issue
Oxybutynin	Non-selective antimuscarinic	Tertiary amine	Only US-approved drug for paediatric non-neurogenic OAB — approval on observational data, no placebo comparison ; liquid available → often first-line; ER more effective and better tolerated than IR	Dry mouth, flushing, constipation, cognitive change ; patch → skin irritation , avoid same site within a week
Propiverine	Non-selective	—	Not available in the US; approved in Canada for paediatric OAB; efficacy similar to oxybutynin with fewer adverse effects	—
Tolterodine	Non-selective; bladder > salivary gland selectivity	—	RCT of 2 mg ER versus placebo: no overall benefit, but children <35 kg benefited → under-dosing of larger children; tolerated after oxybutynin failure or intolerance with similar efficacy	Under-dosing in heavier children
Fesoterodine	Non-selective	Tertiary amine	FOXY randomised crossover versus oxybutynin: similar efficacy, no serious adverse effects	—
Tropium	Non-selective	Quaternary amine — does not cross the blood-brain barrier	Small placebo-controlled RCT: improved OAB symptoms	Empty stomach, 1 hour before food — limits use in children
Solifenacin	Selective M3 > M2	Tertiary amine	LION and LEOPARD phase III multicentre placebo-controlled RCTs: improved all efficacy parameters , mostly mild-moderate side effects	Long half-life ; release not solely coating-dependent → tablet has been cut and started at 2.5 mg
Mirabegron	β_3 -agonist — detrusor relaxation during storage	—	FDA-approved above 3 years for neurogenic OAB, not for non-neurogenic OAB ; phase I: plasma levels in adult range, no adverse effects; RCT versus solifenacin in new OAB: similar efficacy and side effects ; well tolerated combined with anticholinergics in refractory OAB	Off-label in non-neurogenic OAB
Vibegron	β_3 -agonist (newer)	—	Small retrospective series in refractory OAB: most improved clinically and urodynamically; no discontinuation for intolerance	Evidence limited

FDA, United States Food and Drug Administration.

- **Pattern:** antimuscarinics are distinguished by **selectivity** (receptor subtype or tissue) and by **CNS penetration** (tertiary versus quaternary amine); β_3 -agonists avoid the anticholinergic profile.
- **Traps:** oxybutynin's approval rests on no placebo trial; tolterodine's negative trial reflects under-dosing, not inefficacy; mirabegron is approved only for neurogenic OAB in children.

OnabotulinumtoxinA

- Adults: **intradetrusor onabotulinumtoxinA** supported by moderate-quality evidence for carefully selected OAB; **adult AUA dose 100 units**.
- Children with refractory idiopathic OAB/non-neurogenic detrusor overactivity: **less but promising evidence**, retrospective series.
- **Ghent University** uses **100 units as standard of care** for therapy-resistant OAB since its first report of success in this population; retrospective series aged 4–18 years: complete response **best in enuresis, then daytime incontinence, lowest with both**; retention needing temporary CIC exceptional; most needed a single injection; **mean duration of effect about 12 months**.

INTRADETRUSOR DOSE IN NON-NEUROGENIC OAB

Low dose	High dose
5 units/kg, maximum 150 units	10 units/kg, maximum 300 units
No difference in response; higher PVR with high dose	

Limited data → **5 units/kg, maximum 100 units** is appropriate for refractory paediatric OAB.

Electro-neurostimulation

- Growing evidence in childhood OAB; non-invasive or invasive stimulation of nerves that modulate LUT function; **mechanism unclear**.

NEUROMODULATION FOR PAEDIATRIC OAB

Modality	Technique	Trials and comparators	Verdict
Parasacral transcutaneous electrical nerve stimulation (TENS)	Surface electrodes over S2–S3 foramina	vs sham scapular TENS: high cure; vs oxybutynin: similar questionnaire gains, more complete cure, fewer side effects, less discontinuation ; vs urotherapy alone: no added benefit ; vs parasacral sham with inactivated stimulators: no difference	Varied efficacy; true sham questions a placebo effect in earlier trials
Transcutaneous tibial nerve stimulation (TTNS)	Surface stimulation of posterior tibial nerve → detrusor inhibition	Two sham-controlled RCTs in refractory OAB	Conflicting ; the clinically negative trial still showed larger maximum cystometric capacity and volume at onset of detrusor overactivity
Percutaneous tibial nerve stimulation (PTNS)	34-gauge needle ~5 cm cephalad to the medial malleolus ; correct placement → great toe flexion	Adult AUA: third-line , low-grade evidence; paediatric series heterogeneous — early improvement common, few complete responses at 1 year; versus parasacral TENS, TENS gave far more complete resolution	Weaker than parasacral TENS in children
Sacral neuromodulation	Implanted	Adults: third-line , FDA-approved; children: off-label ; RCTs mainly neurogenic; non-neurogenic series mixed OAB, poor emptying and bowel dysfunction	Complications: lead migration, device malfunction, wound infection, revision

Concept — Paediatric OAB is a diagnosis made after urotherapy and bowel management have been given a fair trial; escalation runs from oral drugs to botulinum toxin or neurostimulation, but each step rests on thinner evidence than in adults, and true-sham data temper enthusiasm for TENS.

Giggle Incontinence

- **ICCS: extensive emptying or leakage during or immediately after laughing; LUT function is normal in the absence of laughter.**
- Distinguish from laughter-exacerbated incontinence in **OAB, urethrovaginal reflux or voiding postponement.**
- Pathophysiology poorly understood; successful treatments target both central and LUT origins.

GIGGLE INCONTINENCE: METHYLPHENIDATE VERSUS BIOFEEDBACK

Methylphenidate	Biofeedback (pelvic muscle exercises)
CNS stimulant (ADHD, narcolepsy); small observational series with high success	Teaches pelvic floor contraction and relaxation; small series with high success
Irritability, agitation, sleep disturbance; relapse after stopping	More durable long-term response

Underactive Bladder

- **ICCS: children who need to raise intra-abdominal pressure to initiate, maintain or complete voiding.**
- Pressure-flow studies struggle to separate **detrusor underactivity from outlet obstruction** → trials mix detrusor underactivity with dysfunctional voiding.
- Poor emptying → **CIC helps whatever the cause**, but school-age children with **intact sensation may not tolerate it.**
- RCT biofeedback versus standard urotherapy: both improved voiding pattern and UTI rate, **biofeedback more so**; pelvic EMG improved in both → **many had dysfunctional voiding.**
- RCT in **urodynamically confirmed detrusor underactivity**: urotherapy ± **transcutaneous interferential electrical stimulation** (15 sessions) → more voiding episodes, smaller bladder capacity, higher Qmax, shorter voiding time at 1 year.

Vaginal Reflux

- **Urethrovaginal reflux** is a common cause of incontinence in young girls: the **vagina fills retrograde during voiding.**
- History: **small-to-moderate leak immediately after voiding**; sometimes after sneeze or cough.
- Treatment is positional: **hips abducted, lumbar spine straight; cough at the end of voiding** to clear the vagina.

Urinary Incontinence in Adolescent Female Athletes

- **Highly prevalent, highest in high-impact (jumping, tumbling) sports.**
- Isolated sports-related incontinence: most showed **pelvic floor laxity on the filling phase of upright voiding cystourethrogram (VCUG)**; **biofeedback first**, more successful with **minimal bladder descent**; many progressed to **bladder neck elevating surgery.**

KEY POINTS: URINARY INCONTINENCE — DAYTIME URINARY INCONTINENCE

- Specific findings at initial evaluation suggest **non-functional causes.**
- **Standard urotherapy is effective first-line treatment.**
- **Constipation treatment precedes specific urotherapy.**

- Specific urotherapy depends on the type of LUT dysfunction.

Concept — Every child with DUI passes through the same filter — exclude organic disease, treat the bowel, optimise neurodevelopment, deliver standard urotherapy — and only then does the specific diagnosis (dysfunctional voiding, OAB, underactive bladder, giggle incontinence, vaginal reflux) choose the specific therapy.

URINARY INCONTINENCE: ENURESIS

- **Prevalence about 5–10% at 7 years**; falls with age, but children with enuresis persisting beyond 10 years have **significantly more daytime LUTS**.
- Telephone survey: overall prevalence **about 2%**, with **no decline with age in adulthood**.
- **Mechanism: mismatch between nocturnal urine production, nocturnal bladder capacity and arousal from sleep.**
- **Familial**; genome-wide association study: risk loci on **chromosomes 6 and 13**, with candidate genes in sleep, urine production and bladder function.

Nonmonosymptomatic Enuresis

NON-MONOSYMPTOMATIC VERSUS MONOSYMPTOMATIC ENURESIS

Non-monosymptomatic (NMNE)	Monosymptomatic (MNE)
Enuresis with any LUTS	Enuresis without any other LUTS
Treat bothersome daytime LUTS first; no evidence to delay enuresis treatment if daytime symptoms are minor	Active therapy recommended from 6 years (no absolute age)

Monosymptomatic Enuresis

- Enuresis impairs **peer relationships and self-esteem** from 6 to 18 years.
- Families not ready for active therapy still need **education to dispel myths** — enuresis as laziness or a marker of abnormal psychology; in the 1960s enuresis with fire-starting and animal cruelty was posited to predict antisocial predatory behaviour.
- **Ineffective strategies** (prospective cohort): **night waking, bedtime fluid restriction, rewards, showing displeasure, protection pants.**
- **Standard urotherapy** is not shown to be effective for enuresis but is low-risk.
- **Strong evidence supports desmopressin and the enuresis alarm as first-line therapy for MNE.**

FIRST-LINE THERAPY FOR MNE: ENURESIS ALARM VERSUS DESMOPRESSIN

	Enuresis alarm	Desmopressin
Target	Arousal — operant conditioning	Nocturnal urine production — ADH analogue
Mechanics	Urine on sensor triggers alarm — already too late that night ; works over months ; continuous use ; parent must wake the child ; needs a motivated family	Tablets or oral lyophilisate (lyophilisate not available in the US); 0.2–0.4 mg tablet or 120–240 µg lyophilisate , given in the evening about 1 hour before bedtime
Continue / stop	If improving after a few weeks (smaller wet spot, occasional dry nights) continue until 14 consecutive dry nights	Effect is immediate → no role in continuing beyond 1 week if no initial effect ; once effective, use for special nights (e.g. sleep-away camp) or nightly with periodic drug holidays
Response	Success about 50–70% , a large proportion cured; better than no treatment for wet nights and for 14 consecutive dry nights	Rule of thirds : about one-third complete, one-third partial, one-third no response; best in MNE with

Enuresis alarm		Desmopressin
		nocturnal polyuria and normal daytime voided volumes
Durability	Durable ; overlearning may improve cure — after 14 dry nights continue the alarm while drinking one to two glasses of water each evening until another 14 dry nights	Not lasting — effect ends with the drug
Safety	—	Safe long-term, side effects rare; overdose → water intoxication and hyponatraemia → avoid excess evening fluid; a thirsty or dehydrated child after sport may still drink and take it safely

- **Failure of either:** switch to the other, **combine them** (low-quality evidence: fewer wet nights, more achieving 14 consecutive dry nights), or move to second-line therapy.

Re-evaluation before second-line therapy

- Confirm **first-line therapy was used correctly**.
- **Uroflowmetry and PVR** are not needed to work up MNE but help decide whether an anticholinergic can be added.
- **Neurodevelopmental disorders:** enuresis is common in ADHD, ASD and ID but is **not a reason to withhold treatment**; bell-and-pad alarm success was **somewhat lower with a neurodevelopmental disorder, with no difference in relapse** (multicentre retrospective).
- **Sleep:** enuretic children sleep differently (systematic review). **Sleep-disordered breathing** → difficult arousal and inefficient breathing → **atrial natriuretic peptide secretion** → **increased urine output**. Obstructive sleep apnoea (OSA) signs: **heavy snoring, daytime sleepiness, behavioural and learning problems**. Secondary analysis of a multicentre RCT of adenotonsillectomy versus watchful waiting in non-severe OSA: **enuresis odds about twice as high with watchful waiting at 7 months; yet only about half respond to adenotonsillectomy**.
- **Constipation:** impaction may occur **without overt gastrointestinal symptoms** → careful evaluation; treating constipation alone cured a minority of MNE.

Second-line: anticholinergics

- **Anticholinergics are second-line:** adding **oxybutynin** to desmopressin in absent or partial responders improved response in most, **more so without constipation**; solifenacin and tolterodine also studied.
- **Check PVR and constipation before adding; initial response followed by recurrence → suspect constipation.**

Therapy-resistant enuresis

- After combined anticholinergic plus desmopressin, further medical therapy is **trial and error**; **previously failed first-line therapy may later work** as physiology changes; **strongly consider a prophylactic bowel regimen**.

AGENTS FOR THERAPY-RESISTANT ENURESIS

Agent	Evidence	Caution
Imipramine 25–50 mg	With desmopressin, more effective than desmopressin alone for 14 consecutive dry nights	Displaced from first line by cardiotoxicity — fatal in overdose or long QT syndrome ; electrocardiogram (ECG) first if unexplained syncope, palpitations or family history of sudden cardiac death
β3-Agonists (mirabegron, vibegron)	Small studies; successful off-label adjunct	Off-label

Agent	Evidence	Caution
Psychoactive drugs	RCTs: fluoxetine, atomoxetine, reboxetine; non-RCT : clonidine	Most urologists defer prescribing to mental health specialists

KEY POINTS: URINARY INCONTINENCE — ENURESIS

- Enuresis results from a **mismatch between nocturnal urine production, bladder storage capacity and sleep arousal**.
- **Bothersome daytime LUTS are treated before enuresis.**
- **Desmopressin and the enuresis alarm are first-line.**
- Neurodevelopmental disorders, sleep-disordered breathing and constipation affect treatment.

Concept — Enuresis is a three-way mismatch: desmopressin reduces output (immediate, not durable), the alarm trains arousal (slow, durable) and anticholinergics enlarge functional capacity; resistance usually signals an unaddressed comorbidity — daytime LUTS, constipation or sleep-disordered breathing.

FUNCTIONAL CONSTIPATION

- **ESPGHAN/NASPGHAN joint evidence-based recommendations** govern treatment.
- Faecal incontinence is common in functional constipation; **treatment is the same with or without faecal incontinence**.
- **Alarm signs or symptoms → gastroenterology or other appropriate specialty.**
- Diagnosis rests largely on history and examination.
- **Combined LUT and bowel dysfunction → EAU/ESPU: treat the bowel first.**

FIG. 47.3 ALGORITHM FOR TREATMENT OF CONSTIPATION OF INFANTS ≥6 MONTHS OF AGE (modified from Tabbers et al., ESPGHAN/NASPGHAN, 2014)

Functional constipation → **faecal impaction?** → yes: **disimpact with oral or rectal medication** → effective? no → **refer to paediatric gastroenterologist**; yes → next step.

No impaction (or successful disimpaction) → **education, diary, toilet training, oral medication** → review at **2 weeks** → effective? no → **reassess**: re-education, adherence, different dose, different medication, mental health needs, **untreated impaction** → effective? no → refer; yes → **maintenance therapy**.

Maintenance → relapse? yes → back to reassessment; no → **wean and observe** → relapse after weaning → **refer to paediatric gastroenterologist**.

Disimpaction

- **Disimpaction is the first step.** Faecal impaction = an immobile bulk of dry faeces lodged in the lower gastrointestinal tract after chronic constipation.
- Sometimes, not always, palpable as a hard lower abdominal mass; on DRE or radiograph the sign is a **dilated rectum containing a large amount of stool**.
- **Polyethylene glycol (PEG) is the first-choice laxative for disimpaction**: poorly absorbed **osmotic laxative** → softer, more frequent stools; side effects minor — flatulence, bloating, nausea/vomiting, abdominal cramps.
- Two forms: **PEG 3350** (molecular weight 3350 g/mol) and **PEG 4000** (4000 g/mol). In the US PEG 3350 is sold **with electrolytes** (e.g. GoLYTELY) or **without** (e.g. MiraLAX).
- **Renal dysfunction or other water/electrolyte disturbance → prescribe PEG with caution and monitor serum electrolytes.**

DISIMPACTION CHOICES COMPARED

Comparison	Finding	Implication
PEG 3350 with versus without electrolytes (inpatient faecal impaction)	Equally effective ; with-electrolyte form had more adverse events — electrolyte abnormalities, abdominal pain, nausea/vomiting	Electrolyte-free PEG adequate
Daily enemas versus oral PEG (systematic review)	Minimal outcome difference ; oral PEG caused more faecal incontinence	ESPGHAN/NASPGHAN favour PEG for its route ; daily enemas for 3–6 days if PEG unavailable ; with severe backup, enemas (distal to proximal) cause less distension, nausea and emesis

TABLE 47.6 DOSES OF COMMONLY USED LAXATIVES

Laxative	Recommended dose	Age
PEG 3350 — disimpaction	1–1.5 g/kg/day for 3–6 consecutive days	>3 years
PEG 3350 — maintenance	0.2–0.8 g/kg/day in 1–2 doses; starting dose 0.4 g/kg/day	>6 months
Lactulose — maintenance	1–2 g/kg once or twice a day	>6 months

- **Pattern:** the disimpaction dose of PEG is roughly double the top of the maintenance range, given for days rather than months.
- **Trap:** the disimpaction dose is listed for children older than 3 years, whereas both maintenance agents are listed from 6 months.

Maintenance

- After disimpaction, **maintenance therapy prevents re-accumulation; PEG is first-line maintenance; start 0.4 g/kg/day**, then individualise dose and frequency.
- **At least 2 months of maintenance** (low-level evidence); about half of children stop maintenance within 6–12 months (systematic review).
- **Weaning criteria:** symptoms, including associated LUTS, markedly reduced or absent for **at least 1 month; defecation at least three times a week; no longer meeting Rome IV** → reduce dose and frequency gradually and monitor for relapse.
- Relapse on withdrawal → **reinstate until something in the child's life changes** (growth, diet, behaviour); **laxatives do not correct underlying motility, diet, behavioural or biomechanical problems.**

Titration

- **No bowel movement within 24 hours** → dose ineffective; stool is accumulating.

OSMOTIC VERSUS STIMULANT LAXATIVES

Osmotic (PEG, lactulose)	Stimulant (senna, bisacodyl)
Soften stool; dose escalation gives looser, watery stool → more accidents	Switch to a stimulant when osmotic escalation causes accidents; aids ejection of stool
Some children need a stimulant for ejection plus a smaller osmotic dose to keep stool soft	

Safety and dependence

- **PEG 3350:** families reported neurological and behavioural symptoms to the FDA; **NASPGHAN** — **no evidence of serious neurological side effects**; daily use causes **no sustained rise in blood glycol levels**; in animals PEG 3350 alters the microbiome but not behaviour.
- **Lactulose** is the alternative osmotic laxative; Cochrane review: **safe and effective from 6 months of age**. Common side effects of both: diarrhoea, nausea, bloating.

- **No physical or psychological dependence has been reported with PEG 3350 or lactulose; abrupt cessation without weaning precipitates relapse that is mistaken for dependence.**
- Drugs of dependence create a need unrelated to innate dysfunction; **constipation from dysmotility is an innate dysfunction**, and some children simply require laxatives for effective bowel function.

Standard Nonpharmacologic Management

- Disimpaction addresses the acute problem; **education and non-pharmacological measures prevent recurrence.**
- Developmentally appropriate teaching of bowel anatomy and physiology (as in standard urotherapy) and of the behavioural changes needed.
- **Relaxed posture with foot support, knees at or above hip level** → reduces the anorectal angle.
- **Scheduled toilet sitting after meals** exploits the **gastrocolic reflex**.

LIFESTYLE MYTHS VERSUS EVIDENCE (ESPGHAN/NASPGHAN: NORMAL FLUID, NORMAL FIBRE, NORMAL ACTIVITY)

Measure	Evidence	Advice
Extra fluid	Randomised study: raising daily water intake by half changed neither stool frequency nor consistency	Normal intake; excess water does not help
Extra fibre	Adults improve with supplementation above 10 g/day; children: no or only marginal improvement	Fibre is an essential stool " normaliser " (softens hard stool, firms loose stool); target about 10 g/day in young children rising to about 20 g/day in adolescents — most fall short; mix fermentable (fruit, vegetables, grains) and bulking (nuts, seeds, wheat bran, oat bran) sources
Extra activity	No RCTs ; cross-sectional study of Brazilian children aged 6–12: no association with functional constipation	Regular activity as part of a healthy lifestyle
Probiotics	Cochrane: possibly no difference versus placebo, or as an add-on to laxatives	Not recommended routinely

- **Trap:** families escalating fluid, fibre or exercise often do so to **avoid** effective treatment — education on these myths should not delay laxatives.

Refractory Constipation: Nonsurgical Management

- Non-responders to education and disimpaction/maintenance, or recurrent relapse → **access to multidisciplinary paediatric gastroenterology** — further testing, psychological therapy and bowel-focused invasive interventions beyond a urology practice.

NEWER ADULT AGENTS FOR CONSTIPATION

Class	Agents	Action	Paediatric status
Prosecretory	Lubiprostone, linaclotide	Decrease gastrointestinal transit time	No paediatric randomised studies ; may suit some older adolescents
Highly selective serotonergic agonist	Prucalopride	Prokinetic	

- **Psychological therapy:** constipation often stems from **fear and avoidance of painful defecation**; RCT of behavioural therapy plus laxatives versus conventional treatment: **no change in success, but fewer behavioural problems** → bowel-focused psychological intervention helps adherence.

- **Pelvic floor physical therapy** is an effective adjunct.
- **Anorectal manometry biofeedback** teaches relaxation during defecation in **dyssynergic defecation** — pelvic floor contracts instead of relaxing, the bowel counterpart of dysfunctional voiding; **beneficial in adults, inconclusive in children**.
- **Transanal irrigation (TAI)**: effective and accepted in **neurogenic bowel and anorectal malformations**; in functional constipation, retrospective series show efficacy. Rectal catheter or cone → water or saline infused retrograde → evacuation. **Despite high success, about a quarter to a third of families stop** — more often in functional constipation, probably from discomfort during insertion or irrigation.

Refractory Constipation: Surgical Management

- After failure of laxatives and retrograde enemas, subsequent therapies are invasive.
- **Antegrade continence enema (ACE)**: irrigant ± laxative via a **Malone appendicostomy or percutaneous caecostomy tube**, giving regular, simplified colonic evacuation; **no randomised comparison of ACE types**; complications — **leakage, tube dislodgement, stomal stenosis, skin infection**.
- Urologists know ACE from neurogenic bladder/bowel, but in **non-neurogenic BBD, anorectal and colonic manometry guide management**.

Pre-ACE pathway: constipation suggestive of Hirschsprung disease → **anorectal manometry for RAIR** → **absent** → **rectal biopsy** (ESPGHAN/NASPGHAN) → RAIR present but **raised anal sphincter pressure** → **consider botulinum toxin before ACE** → **traditional enemas first** — often succeed, avoiding surgery; ACE reserved for when enemas work and an easier route is wanted → **colonic manometry to assess motility** before ACE.

COLONIC MOTILITY ON MANOMETRY AND SURGICAL CHOICE

Colonic manometry	Expected result
Normal colonic function	Empties well with ACE ; better evacuation than left-sided or pancolonic dysfunction
Segmental dysmotility	Likely to empty well with ACE; multi-institutional Pediatric Colorectal and Pelvic Learning Consortium: ACE flushing effective in most ; a small minority later needed colonic resection — more likely without pelvic floor dyssynergia on anorectal manometry
Diffuse dysmotility	Complex: degree of dysmotility, colonic dilatation and anorectal dynamics decide extent of resection, need for concomitant ACE, and diverting ostomy versus primary anastomosis ; many do better with enemas from below , some need intestinal diversion ; after diversion, manometric return of colonic motility guides timing and type of ostomy closure

- **Trap**: an ACE is an antegrade washout, so it needs a colon that can move the washout through; the worse and more diffuse the dysmotility, the less an ACE alone will achieve.

Refractory Constipation: Additional Therapies

- The 2014 ESPGHAN/NASPGHAN guidelines only briefly addressed botulinum toxin and nerve stimulation, with limited evidence.
- **OnabotulinumtoxinA** (inhibits acetylcholine release → muscle paralysis): internal anal sphincter injection is **safe for internal anal sphincter achalasia**; may ease stool passage in suspected functional outlet obstruction with **high pressures on anorectal manometry**; later studies mixed; in severe constipation it **may work regardless of sphincter dynamics**.

NEUROMODULATION FOR CHILDHOOD CONSTIPATION

Modality	Evidence	Verdict
Implantable sacral nerve stimulation	Systematic review: symptom improvement; high complication rate and most needed further surgery for removal or repositioning ; later studies similar	Effective but reoperation-prone
Transcutaneous sacral / parasacral TENS	Improvement	Reasonable before more invasive therapy
Transcutaneous interferential electrical stimulation	Improvement	
Percutaneous tibial nerve stimulation	Similar results in cohorts with more functional constipation	

In the systematic review these were the only modalities studied in children without anorectal malformation, neurological abnormality or Hirschsprung disease; populations and outcomes were heterogeneous.

Recurrent Urinary Tract Infections

- BBD is common in children with UTI, and recurrent UTI is common in children with BBD.

RIVUR AND CUTIE (COMBINED ANALYSIS)

	RIVUR	CUTIE
Population	Children <6 years after a first or second febrile and/or symptomatic UTI	
Reflux	With VUR	Without VUR
Intervention	Randomised antibiotic prophylaxis versus placebo	
BBD at baseline	More than half of toilet-trained children	
Recurrent UTI	BBD + VUR 51% > BBD alone 35% > VUR alone 20%	
Constipation sub-analysis	Two of 19 sites with scout radiographs at baseline VCUG: wider caecal diameter → more recurrent UTI	

- **Trap: BBD alone carries a higher recurrence risk than VUR alone**; the combination is worst.
- **UTI risk by type of LUT dysfunction: voiding postponement, detrusor underactivity and dysfunctional voiding carry higher UTI rates than detrusor overactivity.**
- LUT dysfunction, especially dysfunctional voiding, can cause **secondary VUR**; **non-surgical treatment of the dysfunction often resolves the reflux.**
- Functional constipation: systematic review reports **high UTI prevalence.**

GUIDELINES ON BBD AND UTI

Body	Recommendation
EAU-ESPU	Strongly recommends excluding BBD in any toilet-trained child with febrile and/or recurrent UTI; BBD + recurrent UTI → antibiotic prophylaxis during the BBD treatment period
ICCS	Prophylaxis usually maintained until LUT dysfunction and constipation improve symptomatically
AUA	Antibiotic prophylaxis for children with BBD and VUR (highest recurrence risk)

Concept — In recurrent UTI the bladder and bowel dysfunction is the modifiable risk factor and outranks reflux itself; treat the dysfunction under prophylactic cover, and reflux secondary to it often resolves.

KEY POINTS: FUNCTIONAL CONSTIPATION

- **Disimpaction with a laxative is the first step.**
- **Maintenance for at least 2 months** prevents faecal re-accumulation.
- **Myths about increased fluid and fibre should not delay more effective treatment.**
- Refractory constipation warrants access to **paediatric gastroenterology**.
- Although urologists are familiar with ACE, **anorectal and colonic manometry often guide surgical management** in this population.

Concept — *Constipation is treated mechanically before it is treated with lifestyle: empty the rectum, keep it empty long enough to recover, then wean slowly; fluid, fibre and exercise are normalised rather than escalated, and refractory disease is mapped by manometry before any operation.*